Type-Aware Failure Prediction for Chart Reasoning in Vision-Language Models

Can a model's internal state tell us *how* it is about to go wrong?

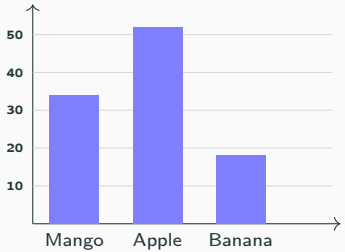
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The problem

Vision-language models read charts, and sometimes get them wrong



A **vision-language model (VLM)** takes an image and a question and writes an answer. We study two open 7B models: **Qwen2.5-VL** and **LLaVA-NeXT**.

Q: "What is the value of Mango?"
Right answer: 34

Q: "What is the value of Guava?"
Right answer: *there is no Guava bar*

Two different ways to be wrong

Misread ("structural")

- "Mango?" → 30
- The answer *was* on the chart; the model read it badly (rounded to a gridline, took the next bar)
- Fix: **look again**, more carefully

Fabrication

- "Guava?" → 40
- The model states something the chart does *not* contain
- Looking again cannot help. Fix: **don't answer**

Today's reliability tools give one "trust / don't trust" score. They cannot tell these apart, so they cannot pick the right fix.

Our question

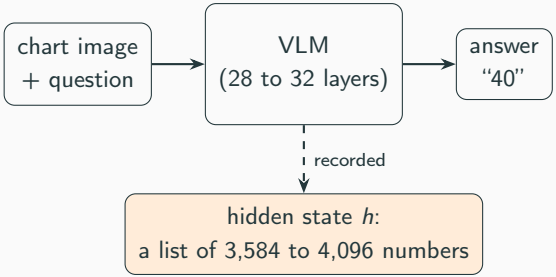
Before the model answers, can we tell from its **internal state** whether it is about to be wrong, and **in which way**?

Hypothesis (type separability). The internal state before a misread differs from the state before a fabrication, because one is a perception error and the other is an answer not grounded in the image.

And if so: does the **type** decide which fix works?

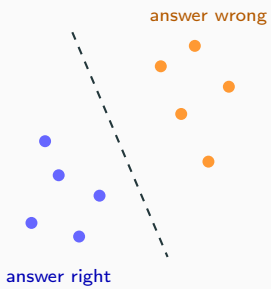
Three tools we use

Tool 1: the hidden state



- Inside, the model turns its input into long vectors of numbers, layer by layer
- We record the vector at the **last word of the question**, the moment just before it writes its first answer word, at every layer
- Recorded in the same run that produced the answer, so it belongs to exactly that answer

Tool 2: a probe



- A **probe** is a simple classifier (logistic regression) trained on hidden states to predict a label, e.g. "will this answer be wrong?"
- It can only draw a straight line through the space
- If it succeeds on charts it never saw, the information is plainly *written in* the model's state

Tool 3: how we score a probe, and why we need baselines

AUROC: pick one example of each class at random; AUROC is the chance the probe scores the right one higher.

$$0.5 = \text{coin flip} \qquad 0.8 = \text{useful} \qquad 1.0 = \text{perfect}$$

Baselines. A probe can "succeed" by reading something trivial, such as how long the question is. So every probe is compared with:

- the same classifier on **surface features** only (question type and length, image size, numeric answer), with no access to the model
- the probe after those features are **removed** from the hidden state
- probes trained **within one question type**

Every number has a 95% confidence interval, and no chart is in both training and test.

Part 1: real charts (ChartQA)

What errors do models actually make on real charts?

Error label	Qwen	LLaVA
Misread	71	583
Arithmetic slip	93	435
Not actually wrong	112	69
Unclear	40	74
Fabrication	**1**	**17**
All errors	317	1,178

- 2,500 ChartQA questions: Qwen 87% right, LLaVA 53%
- We labelled every wrong answer (LLM annotator, written rubric)
- **Fabrication almost never happens** on real charts: questions are about things the chart shows
- 35% of Qwen's "errors" are actually right (format, wrong gold)
- Human check: two of us labelled 200 errors; after settling disagreements, humans and LLM agree at $\kappa = 0.61$

Result 1: we can predict a wrong answer before it is written

AUROC (wrong vs. right)	Probe	Surface features only	Features removed
Qwen	**0.88**	0.72	0.80
LLaVA	**0.80**	0.61	0.78

- Far above what the question's surface reveals, and most of it survives removing those features
- The signal sits in the question-side state in the last third of the network; the image part alone stays near the surface baseline
- Stable across five different train/test splits (± 0.02)

Result 2: misreads and arithmetic slips look different inside

AUROC	Qwen	LLaVA
Misread vs. correct	0.89	0.81
Arithmetic slip vs. correct	0.90	0.95
Misread vs. arithmetic slip	**0.90**	**0.96**

- Worry: arithmetic slips only happen on "what is the difference…" questions, so a probe might just spot the wording
- Check: retrain using only arithmetic questions (0.85 and 0.84), or only reading questions (0.91 and 0.81). The signal survives
- On real charts, the two error types models actually make are separable

Part 2: making fabrication happen

Synthetic charts: ask about something that is not there

We drew 900 bar charts and asked each the same question twice, **in identical words**: once about a bar that is shown, once about one that is not.

- Pilot 1: "What is the value of Mango?" vs. "... of Guava?"
- Pilot 2: lookalike names (Dublin shown, ask Dubai); also "Which is larger?" and "Which bar is next to...?"
- Pilot 3: years 2014 to 2019 shown, ask about 2021

Identical wording matters: the question text alone cannot tell which case it is, so a probe cannot cheat by reading the words.

The models almost never say “it’s not there”

Asked for a missing bar’s value (600 questions)	Qwen	LLaVA
Says it is not shown	**15**	**0**
Says “0”	494	232
Makes up a number	91	368

- Qwen mostly says “0”; when it invents, it usually copies the lookalike bar (asked Dubai, reports Dublin)
- LLaVA invents a number for most missing bars

. . . but inside, they know the bar is missing

AUROC, missing vs. shown name	Qwen	LLaVA
Probe on hidden state	**0.99 to 1.00**	**0.96 to 1.00**
Classifier on the question text only	0.49 to 0.52	0.49 to 0.52
Probe on the image part only	0.50	0.50

- The models register that the asked bar is missing, **then answer anyway**
- The controls are at chance: it is neither the words nor the image alone, but the model combining the two
- Qwen knows by layer 2 of 28; LLaVA by about layer 9 of 32

Is it really about "missing", or about "can't be answered"?

Pilot 3: the chart shows 2014 to 2019; we ask about 2021. Nothing is "missing", the question just cannot be answered from the chart.

Answer to 2021	Qwen	LLaVA
Says not shown	**102**	0
"0"	76	19
Makes up a number	105	281

- The probe trained on missing *names* also flags out-of-range *years*: **1.00** (Qwen), **0.91** (LLaVA)
- It fires whether Qwen then refuses or invents: it tracks "the chart can't answer this", not what the model will say

Misread vs. fabrication: does each probe stick to its own type?

	Qwen		LLaVA	
Scored on →	misreads	fabrications	misreads	fabrications
Misread probe	**0.89**	0.59	**0.78**	0.88
Fabrication probe	0.58	**1.00**	0.68	**0.99**

- **Qwen:** each probe is strong on its own type, weak on the other: separable, as hypothesised
- **LLaVA:** the misread probe fires even more on fabrications: one general "not grounded" signal
- **Caveat:** on synthetic charts, whether a reading counts as a misread depends mostly on where the value sits between gridlines; that chart property alone predicts misreads as well as the probes (0.92 and 0.86)

Part 3: does the type decide the fix?

Four ways to make the model look again

- **Re-ask**: same question again (a second try)
- **Upsample**: the whole chart, enlarged
- **Crop**: only the asked bar and its axis, enlarged
- **Verify**: "first name the part of the chart that supports your answer"

Prediction, written down before running (pre-registered): cropping should rescue misreads more than fabrications, since for a fabrication there is nothing to look at.

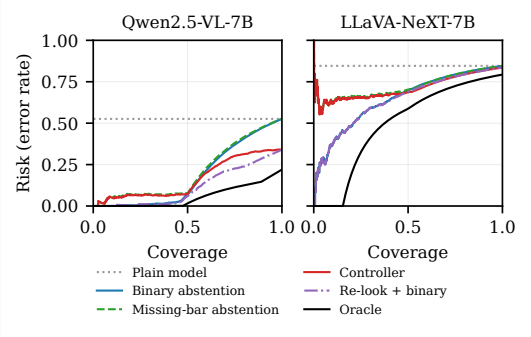
Applied to every wrong answer on the synthetic charts (Qwen 631, LLaVA 1,015).

Result: the prediction holds

Fraction fixed	Re-ask	Upsample	Crop	Verify
Qwen				
Misread	0.11	0.13	**0.26**	0.09
Made-up number	0.00	0.00	0.00	**0.14**
"0"	0.02	0.02	0.02	**0.69**
LLaVA				
Misread	0.03	0.01	**0.15**	0.03
Made-up number	0.00	0.00	0.00	0.00

- Crop helps misreads, never fabrications, in both models (pre-registered test passes: +0.26 and +0.15, both intervals above zero)
- For Qwen the best fix differs by type: crop for misreads, verify for missing bars (it makes Qwen say "not shown")
- Nothing we tried fixes LLaVA's missing-bar answers: only abstaining works

Putting it together: a type-aware controller



Controller: if the missing-bar probe fires, abstain; if the misread probe fires, look again with the best fix.

Risk-coverage: answer only the questions you are most confident about; plot the error rate against the fraction answered. Lower is better.

- Re-looking lowers error in both models
- The controller beats plain "abstain when unsure" only for Qwen, mostly by turning "0" into "not shown"
- A perfect-knowledge oracle is still far better: room to improve

Conclusions

What we learned

1. A model's internal state predicts that an answer will be wrong **before** it is written, beyond the question's surface
2. On real charts the errors are misreads and arithmetic slips, and they look **different** inside the model
3. Models **know** when a chart cannot answer the question, and answer anyway. This "unanswerable" signal is the most useful one for deciding when to abstain
4. The type does change the best fix: cropping rescues misreads, not fabrications, as we predicted in advance

Limitations

- Fabrication is rare on real charts, so the misread vs. fabrication comparison rests on synthetic bar charts
- On synthetic charts, misreads largely reflect chart geometry, and those probes do not carry over to real charts
- Error labels come from an LLM annotator; two humans agreed with each other only moderately ($\kappa = 0.42$)
- Repairs and the controller are tested on synthetic charts only
- Probes show what is encoded, not that the model uses it

Thank you

Code and results: github.com/the-neemon/type-aware-vlm-failures